

Hyunkyung Han

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Seoul, South Korea

RESEARCH INTERESTS

Computer Vision, Generative Models, Lightweight Models, Medical Imaging, Image-to-Image Translation, Image Registration, Domain Generalization

EDUCATION

- **Yonsei University** Mar 2025 - Present
M.S. in Artificial Intelligence
◦ Advisor: Seong Jae Hwang Seoul, South Korea
- **Ewha Womans University** Mar 2021 - Feb 2025
B.S. in Computer Science and Engineering Seoul, South Korea

CORPORATE EXPERIENCE

- **NewcureM** Feb 2023 - Aug 2025
AI/SW Team Researcher Seoul, South Korea
 - Radiation Dose/Time Reduction Solution for Brain PET (FBB, FMM, 18F-FDG, 18F-FPCIT)
 - AI-guided Prostate Lesion Detection Solution for PSMA-1007 PET/CT and T2, ADC, DWI MRIs
 - Patient-specific Diagnosis Report Generation (Brain, Prostate)
 - K-FDA Approvals for Medical SW Devices

PROJECTS

- **Lightweighting Diffusion Transformers on Mobile Devices** Mar 2026 – Present
Samsung Electronics
 - Hardware-faithful token merging method for diffusion transformers for image generation on mobile devices

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION, T=THESIS

- [C.1] K. Choo, Y. Kim, **Hyunkyung Han**, G. Park, C. Kim, S. Jung, S. J. Hwang. **QWERTY: Training-Free Motion Control via Query-Warped Video Diffusion Transformers**, *European Conference on Computer Vision (ECCV)*, 2026
- [C.2] K. Choo, **Hyunkyung Han**, J. Kim, C. Yoon, S. J. Hwang. **Mono-Modalizing Extremely Heterogeneous Multi-Modal Medical Image Registration**, *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025
- [C.3] **Hyunkyung Han**, S. Lee, K. Choo, D. Kim, H. Lim, D. Kim, S. B. Seo, Y. Yoon, M. Yun. **Denoising for 1/4 Count PET Scans with Dose and Scan Time Reduction**, *American College of Nuclear Medicine (ACNM), Annual Meeting*, 2025
- [C.4] S. Cho, **Hyunkyung Han**, S. Lee, M. Jung, Y. Oh, M. Yun. **Assessment of Tumor Response to Neoadjuvant Hormone Therapy in Prostate Cancer Using F-18 PSMA-1007 PET: A Feasibility Study**, *European Association of Nuclear Medicine (EANM)*, 2025
- [C.5] S. Seo, M. So, S. Lee, Y. Yoon, D. Seo, D. Kim, **Hyunkyung Han**, S. Kang, D. Kim, M. Yun. **Stratified Evaluation of Deep Learning Models by Age and Cognition Level for Amyloid Positivity Prediction in 18F-FDG PET**, *European Association of Nuclear Medicine (EANM)*, 2025
- [C.6] M. So, S. Seo, S. Lee, H. Lim, Y. Yoon, D. Kim, **Hyunkyung Han**, D. Seo, K. Choo, Y. Jun, S. Kang, D. Kim, Y. Lee, M. Yun. **Periventricular White Matter Hyperintensities and Brain Metabolism Contribute to Cognitive Dysfunction in Amyloid-Negative Patients**, *European Association of Nuclear Medicine (EANM)*, 2025
- [C.7] **Hyunkyung Han**, K. Choo, S. Kang, H. Lim, M. Yun. **Scan-time Reduction of Brain 18F-FP-CIT PET using DCLGAN**, *Society of Nuclear Medicine and Molecular Imaging (SNMMI)*, 2024
- [C.8] D. Kim, S. Lee, D. Kim, K. Choo, H. Lim, **Hyunkyung Han**, J. Lee, M. Yun. **Deep-learning based multi-classification of FDG PET/CT images for the diagnosis of neurodegenerative diseases**, *European Association of Nuclear Medicine (EANM)*, 2024
- [C.9] D. Kim, S. Lee, K. Choo, H. Lim, **Hyunkyung Han**, J. Lee, S. Seo, S. Baek, C. Yu, M. Yun, J. Yang. **CT-based deep learning framework for analyzing brain PET/CT**, *IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and Room Temperature Semiconductor Detector Conference (RTSD)*, 2024

- [J.1] S. Cho, H.J. Seo, **Hyunkyung Han**, T. J. Jeon, Y.T. Oh, M.Jung, M. Yun. **The Feasibility of [18F]PSMA-1007 PET/CT for Response Assessment in Prostate Cancer Following Neoadjuvant Androgen Deprivation Therapy Compared with mpMRI**, *Cancer Imaging*, 2026
- [J.2] M. So, S. Seo, D. Kim, S. Lee, H. Lim, Y. J. Yoon, D. Kim, **Hyunkyung Han**, Y. Jun, S. Kang, H. J. Seo, T. J. Jeon, S. Cho, H. J. Yoon, Y. Lee, M. Yun. **Association of White Matter Hyperintensities, Regional Brain Glucose Metabolism, and Cognitive Impairment in A β -Negative Patients**, *Neurology*, 2026
- [J.3] **Hyunkyung Han**, K. Choo, T. J. Jeon, S. Lee, S. Seo, D. Kim, S.J. Kim, S. H. Lee, M. Yun. **GAN-based Denoising for Scan Time Reduction and Motion Correction of 18F FP-CIT PET/CT: A Multicenter External Validation Study**, *Clinical Nuclear Medicine*, 2025 (Impact Factor = 9.6)
- [J.4] Y. J. Yoon, S. Seo, S. Lee, H. Lim, K. Choo, D. Kim, **Hyunkyung Han**, M. So, H. Kang, S. Kang, D.Kim, Y. Lee, D. Shin, T. J. Jeon, M. Yun. **Centiloid values from deep learning-based CT parcellation: a valid alternative to freesurfer**, *Alzheimer's Research & Therapy*, 2025
- [J.5] S. Seo, Y. J. Yoon, S. Lee, H. Lim, K. Choo, D. Kim, **Hyunkyung Han**, S. Kang, J. Park, P. H. Lee, D. Kim, M. Yun. **Striatal dopamine transporter uptake predicts neuronal hypometabolism and visuospatial function in Parkinson's disease**, *European Journal of Nuclear Medicine and Molecular Imaging (EJNMMI)*, 2025

PATENTS

- [1] Hyunkyung Han, **Method and Device for Preprocessing Medical Images**.
Korean Patent Application No. 10-2025-0198634, filed Dec. 2025. (Pending)

SKILLS

Python, PyTorch, Python Libraries for Medical Image (Nibabel, MONAI, Scikit-Learn, etc.) Git, Docker, AWS, Linux
DICOM protocol, PACS

ACADEMIC SERVICE

- **Conference/Journal Reviewer**
 - MICCAI 2026
 - MICCAI Workshop MSB Emerge 2025
- **Teaching Assistant**
 - Problem Solving and SW Programming, Ewha Womans University 2023

SCHOLARSHIPS

- AI Seoul Tech Research Program (20M KRW) Seoul Future Foundation, 2026